# ENSAYOS NO DESTRUCTIVOS (END)

## PRACTICA METODOS SUPERFICIALES (PM y LP)

# LÍQUIDOS PENETRANTES

Pieza a ensayar: Soporte eje motriz.

Norma de ensayo: ASTM E 165

Técnica de ensayo: Líquidos penetrantes visibles, removible con solvente. Revelador Húmedo no acuoso

Personal certificado según Norma IRAM NM ISO 9712

Designation: E 165 – 09

# Standard Practice for Liquid Penetrant Examination for General Industry[1]

This standard is issued under the fixed designation E 165; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ε) indicates an editorial change since the last revision or reapproval.

## 1. Scope

1.1 This practice[2] covers procedures for penetrant examination of materials. Penetrant testing is a nondestructive testing method for detecting discontinuities that are open to the surface such as cracks, seams, laps, cold shuts, shrinkage, laminations, through leaks, or lack of fusion and is applicable to in-process, final, and maintenance testing. It can be effectively used in the examination of nonporous, metallic materials, ferrous and nonferrous metals, and of nonmetallic materials such as nonporous glazed or fully densified ceramics, as well as certain nonporous plastics, and glass.

1.2 This practice also provides a reference:

1.2.1 By which a liquid penetrant examination process

*priate safety and health practices and determine the applicability of regulatory limitations prior to use.*

## 2. Referenced Documents

2.1 *ASTM Standards:*[3]

D 129 Test Method for Sulfur in Petroleum Products (General Bomb Method)

E 516 Practice for Testing Thermal Conductivity Detectors Used in Gas Chromatography

D 808 Test Method for Chlorine in New and Used Petroleum Products (Bomb Method)

D 1193 Specification for Reagent Water

D 1552 Test Method for Sulfur in Petroleum Products (High Temperature Method)

# NORMA DE REFERENCIA

# LIQUIDOS PENETRANTES
(TÉCNICAS)

| TABLE 1 Classification of Penetrant Testing Types and Methods |
| --- |
| Type I—Fluorescent Penetrant Testing |
| Method A—Water-washable (see Practice E1209) |
| Method B—Post-emulsifiable, lipophilic (see Practice E1208) |
| Method C—Solvent removable (see Practice E1219) |
| Method D—Post-emulsifiable, hydrophilic (see Practice E1210) |
| Type II—Visible Penetrant Testing |
| Method A—Water-washable (see Practice E1418) |
| Method C—Solvent removable (see Practice E1220) |

Preparación de la superficie

Aplicación del penetrante

- Aerosol
- Aspersión
- Inmersión
- Pincel

- Coloreados o visibles
- Fluorescentes
- Duales

Remoción del exceso

- Con agua
- Con solvente

- Lavables con agua
- Removibles con solvente
- Post emulsificables

Aplicación del revelador

- Aerosol
- Aspersión
- Inmersión
- Espolvoreo

- Seco
- Húmedo acuoso
- Húmedo no acuoso

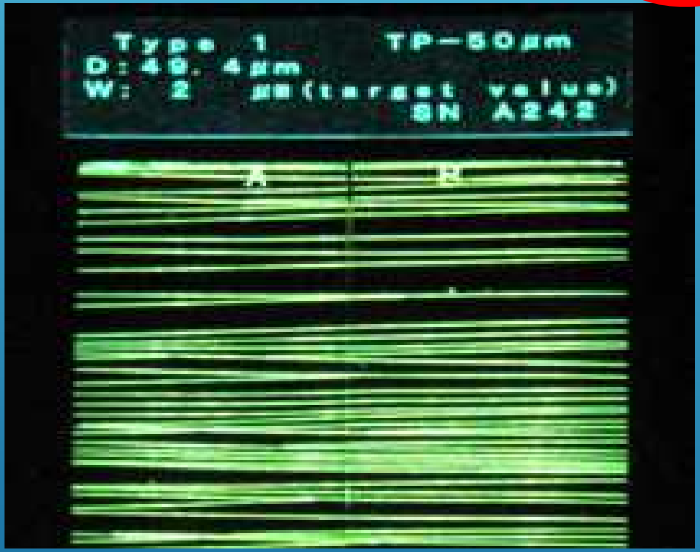
Observación y registro

Debe verificarse su calidad y sensibilidad



# INSUMOS

- Juego de dos paneles (100 x 35mm) con el mismo patron de fisuras. Permite comparar dos sistemas penetrantes simultaneamente. Tamaño de las fisuras: 10µm, 20µm, 30µm, 50µm.




PATRON UTILIZADO

# PATRONES DE SENSIBILIDAD

# PREPARACIÓN DE LA SUPERFICIE

- Superficie en muy buena condiciones
- Desengrasado con solvente

surfaces in the as-welded, as-rolled, as-cast, as-forged, or ceramics in the densified condition.

8.3.1 *Surface Treatment*—Final penetrant examination may be performed prior to treatments that can smear the surface but not by themselves cause surface discontinuities. Such treatments include, but are not limited to, vapor blasting, deburring, sanding, buffing, sandblasting, or lapping. Performance of final penetrant examination after such surface treatments necessitates that the part(s) be etched to remove smeared metal from the surface prior to testing unless otherwise agreed by the contracting parties. Note that final penetrant examination shall always precede surface peening.

NOTE 7—Sand or shot blasting can close discontinuities so extreme care should be taken to avoid masking discontinuities. Under certain circumstances, however, grit blasting with certain air pressures and/or mediums may be acceptable without subsequent etching when agreed by the contracting parties.



mended because of the potential for damage.

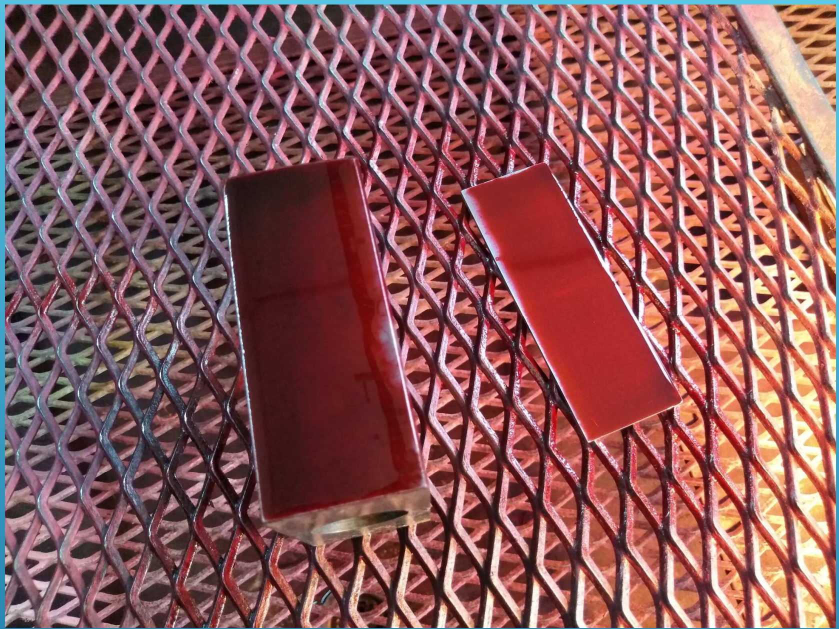
8.4 *Precleaning*—The success of any penetrant examination procedure is greatly dependent upon the surrounding surface and discontinuity being free of any contaminant (solid or liquid) that might interfere with the penetrant process. All parts or areas of parts to be examined must be clean and dry before the penetrant is applied. If only a section of a part, such as a weld, including the heat affected zone is to be examined, all contaminants shall be removed from the area being examined as defined by the contracting parties. "Clean" is intended to mean that the surface must be free of rust, scale, welding flux, weld spatter, grease, paint, oily films, dirt, and so forth, that might interfere with the penetrant process. All of these contaminants can prevent the penetrant from entering discontinuities (see Annex on Cleaning of Parts and Materials).

8.4.1 *Drying after Cleaning*—It is essential that the surface of parts be thoroughly dry after cleaning, since any liquid residue will hinder the entrance of the penetrant. Drying may be accomplished by warming the parts in drying ovens, with infrared lamps, forced hot air, or exposure to ambient temperature.

NOTE 9—Residues from cleaning processes such as strong alkalies, pickling solutions and chromates, in particular, may adversely react with the penetrant and reduce its sensitivity and performance.

8.5 *Penetrant Application*—After the part has been cleaned

Norma de ensayo ASTM E 165



Se aplico el liquido penetrante

Tiempo de penetración: 10 min

**TABLE 2 Recommended Minimum Dwell Times**

| Material | Form | Type of Discontinuity | Dwell Times[A] (minutes) | |
|---|---|---|---|---|
| | | | Penetrant[B] | Developer[C] |
| Aluminum, magnesium, steel, brass and bronze, titanium and high-temperature alloys | castings and welds | cold shuts, porosity, lack of fusion, cracks (all forms) | 5 | 10 |
| | wrought materials—extrusions, forgings, plate | laps, cracks (all forms) | 10 | 10 |
| Carbide-tipped tools | | lack of fusion, porosity, cracks | 5 | 10 |
| Plastic | all forms | cracks | 5 | 10 |
| Glass | all forms | cracks | 5 | 10 |
| Ceramic | all forms | cracks, porosity | 5 | 10 |

[A] For temperature range from 50° to 125 °F [10° to 52 °C]. For temperatures between 40° and 50 °F [4.4° and 10 °C], recommend a minimum dwell time of 20 min.
[B] Maximum penetrant dwell time in accordance with 8.5.1.
[C] Development time begins as soon as wet developer coating has dried on surface of parts (recommended minimum). Maximum development time in accordance with 8.8.5.

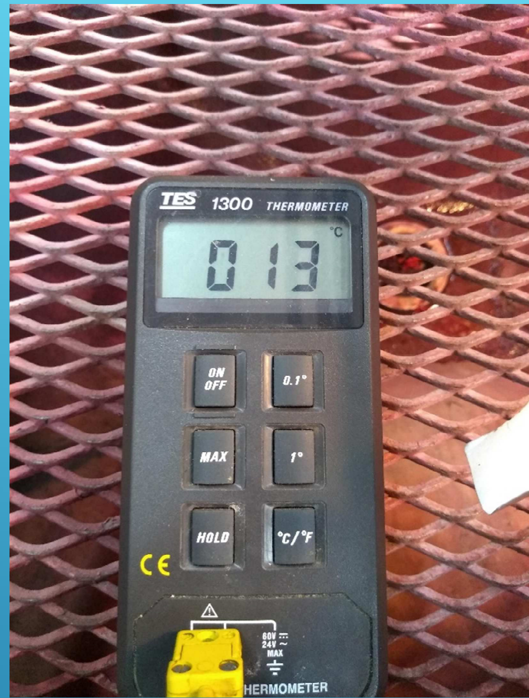
# APLICACIÓN DEL LIQUIDO


Nivel de iluminación

Mayor a 1076 lux
(8.9.2 ASTM E 165)

Temperatura

Entre 4°C y 52°C
(8.2 ASTM E 165)




# CONTROL DE PARAMETROS

Remoción con solvente

Trapeado

Papel

Bajo condiciones de iluminación adecuadas

ETAPA CRITICA

# REMOCIÓN DEL EXCESO

Revelador húmedo no acuoso

Aplicación: Aerosol


# APLICACIÓN DEL REVELADOR Y OBSERVACIÓN

EVALUACIÓN

**TABLE 2 Recommended Minimum Dwell Times**

| Material | Form | Type of Discontinuity | Dwell Times[A] (minutes) | |
|---|---|---|---|---|
| | | | Penetrant[B] | Developer[C] |
| Aluminum, magnesium, steel, brass and bronze, titanium and high-temperature alloys | castings and welds | cold shuts, porosity, lack of fusion, cracks (all forms) | 5 | 10 |
| | wrought materials—extrusions, forgings, plate | laps, cracks (all forms) | 10 | 10 |
| Carbide-tipped tools | | lack of fusion, porosity, cracks | 5 | 10 |
| Plastic | all forms | cracks | 5 | 10 |
| Glass | all forms | cracks | 5 | 10 |
| Ceramic | all forms | cracks, porosity | 5 | 10 |

[A] For temperature range from 50° to 125 °F [10° to 52 °C]. For temperatures between 40° and 50 °F [4.4° and 10 °C], recommend a minimum dwell time of 20 min.
[B] Maximum penetrant dwell time in accordance with 8.5.1.
[C] Development time begins as soon as wet developer coating has dried on surface of parts (recommended minimum). Maximum development time in accordance with 8.8.5.

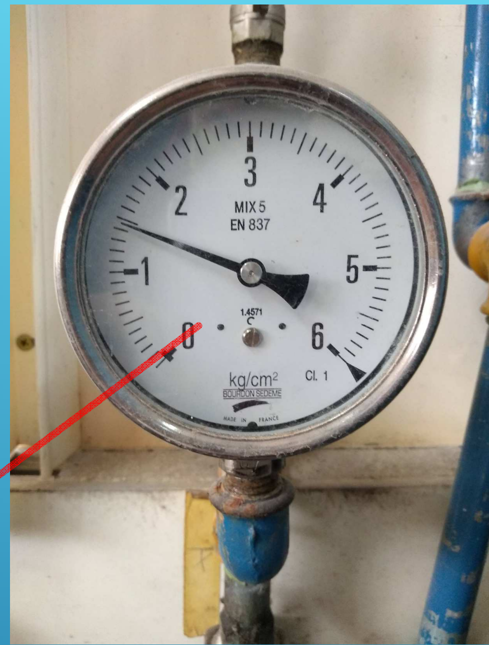
Tiempo de revelado: 10 min

El informe deberá contener toda la información relevante para asegurar la trazabilidad a todos los parámetros y Normas involucradas.

# TIEMPOS DE REVELADO E INFORME


Luz UV
(1000 µWatt/$cm^2$
ASTM E165 8.9.1.1)

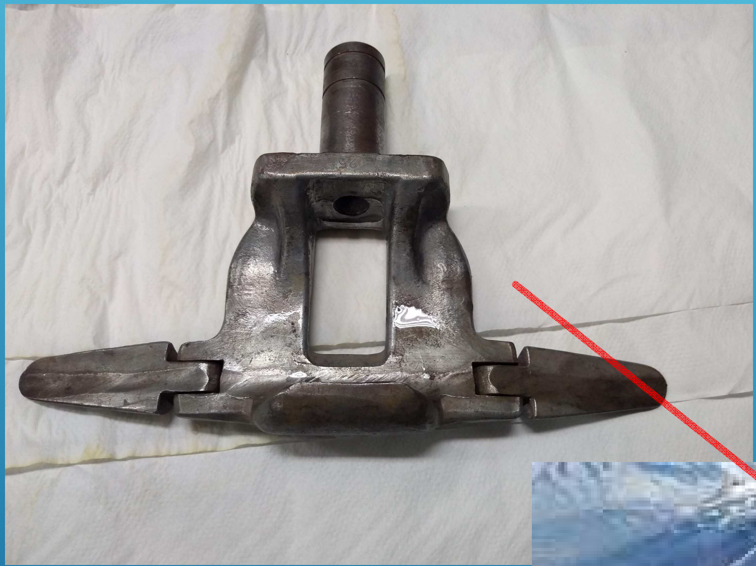
Concentración
emulsificador




Presión agua
(Máx. 40 psi ASTM E165
8.6.1.1 b)

# OTROS PARÁMETROS A CONTROLAR

# PARTÍCULAS MAGNETIZABLES



Pieza a ensayar: Mordaza fija de medio de elevación

Norma de ensayo: ASTM E 3024

Técnica de ensayo: Partículas magnetizables fluorescentes en agua (Magnaflux 20B), método continuo, Yugo (Magnaflux Y1) y C.A.

Personal certificado según Norma IRAM NM ISO 9712

This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.

INTERNATIONAL

**Designation: E3024/E3024M – 19**

## Standard Practice for Magnetic Particle Testing for General Industry[1]

This standard is issued under the fixed designation E3024/E3024M; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ε) indicates an editorial change since the last revision or reapproval.

### 1. Scope

1.1 This practice establishes minimum requirements for magnetic particle testing used for the detection of surface or slightly subsurface discontinuities in ferromagnetic material. This standard is intended for industrial applications. Refer to Practice E1444/E1444M for aerospace applications. Guide E709 may be used in conjunction with this practice as a tutorial.

1.2 The magnetic particle testing method is used to detect cracks, laps, seams, inclusions, and other discontinuities on or

### 2. Referenced Documents

2.1 The following documents form a part of this standard practice to the extent specified herein.

2.2 *ASTM Standards:*[2]

E543 Specification for Agencies Performing Nondestructive Testing

E709 Guide for Magnetic Particle Testing

E1316 Terminology for Nondestructive Examinations

E1444/E1444M Practice for Magnetic Particle Testing

E2297 Guide for Use of UV-A and Visible Light Sources and

# NORMA DE REFERENCIA

# PARTICULAS MAGNETIZABLES (TÉCNICAS)

Preparación de la superficie

↓

Generación del campo magnético

↓

Aplicación de las partículas

↓

Observación y Registro

↓

Desmagnetización

Imán permanente
Circulación corriente
Bobina
Yugo

Corriente Continua
Corriente Alterna

Aerosol
Aspersión

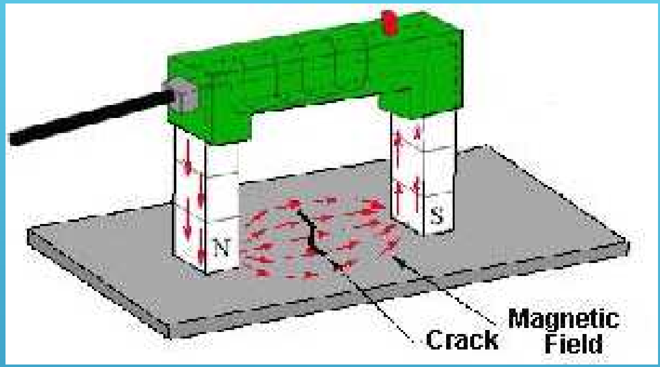
Secas
Vía Húmeda

Coloreados o visibles
Fluorescentes

- **CORRIENTE ELECTRICA (Sistema de magnetización)**
  - Sistema de magnetización donde el campo magnético utilizado para la magnetización de la muestra de ensayo será generado mediante la utilización de corriente eléctrica.
- **CONTINUO (Método de magnetización)**
  - Método de magnetización en el cual las partículas magnetizables son aplicadas cuando el campo magnético generado por la circulación de la corriente de magnetización se encuentra presente en la muestra de ensayo.
- **INDIRECTA (Técnica de magnetización)**
  - Técnica de magnetización mediante la cual la corriente de magnetización no se hace pasar a través de la muestra de ensayo.

# CARACTERÍSTICAS DE LA MAGNETIZACIÓN

- Yugo





Pesas para control de yugo

Según lo establecido por 7.4.4 ASTM E 3024

**FUERZA PORTANTE**

| CORRIENTE | SEPARACION ENTRE PATAS DEL YUGO | |
|---|---|---|
| | 50 a 100 mm (2 a 4 pulgadas) | 100 a 150 mm (4 a 6 pulgadas) |
| ALTERNA | 4,5 Kg (10 lb) | ----- |
| CONTINUA | 13,5 Kg (30 lb) | 23 Kg (50 lb) |

# CONTROL DE LA FUERZA PORTANTE

Nivel de iluminación
Menor a 22 lux
(7.3.1.2 ASTM E 3024)

Luz UV
Mínimo 1000 µWatt/$cm^2$
(7.3.2 ASTM E 3024)

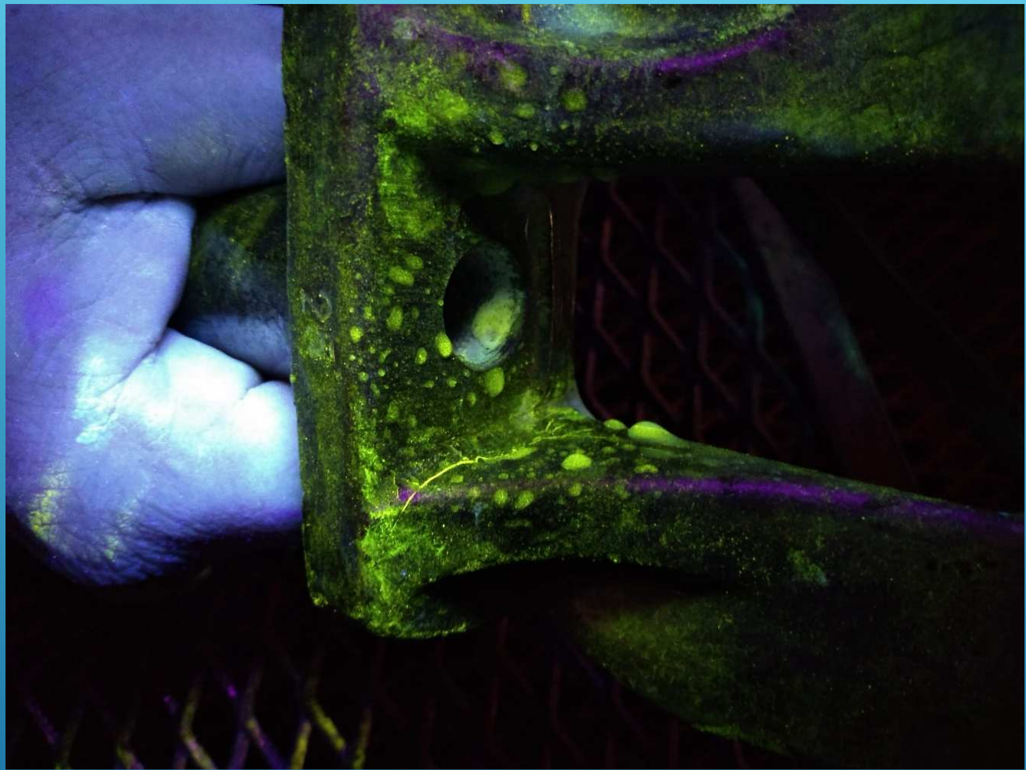

# PARÁMETROS DE ILUMINACIÓN

Toma de muestra del baño
100 ml

Se deja decantar 30 min
(7.2.1.1 ASTM E 3024)

Se mide la concentración
0,1 a 0,4 ml
(5.5.5 ASTM E 3024)

# CONTROL DE CONCENTRACIÓN



# OBSERVACIÓN Y EVALUACIÓN

## Diferencias de criterio

specified. Reference Photographs of indications are noted in E 433).

8.10 *Post Cleaning*—Post cleaning is necessary when residual penetrant or developer could interfere with subsequent processing or with service requirements. It is particularly important where residual penetrant testing materials might combine with other factors in service to produce corrosion and prior to vapor degreasing or heat treating the part as these processes can bake the developer onto the part. A suitable technique, such as a simple water rinse, water spray, machine

and processing techniques may be required. Such examination requires qualification in accordance with 10.2 and the manufacturer's recommendations shall be observed.

**10. Qualification and Requalification**

10.1 *Personnel Qualification*—When required by the customer, all penetrant testing personnel shall be qualified/certified in accordance with a written procedure conforming to the applicable edition of recommended Practice SNT-TC-1A, ANSI/ASNT CP-189, NAS-410, or MIL-STD-410.





10.2 *Procedure Qualification*—Qualification of procedures using times, conditions, or materials differing from those specified in this general practice or for new materials may be performed by any of several methods and should be agreed upon by the contracting parties. A test piece containing one or

used to perform the examination, the agenc… requirements of Practice E 543.

10.4 *Requalification* may be required … substitution is made in the type of penetrant … procedure (see 10.2).



AMS 3045 Magnetic Particles, Fluorescent, Wet Method, Oil Vehicle, Ready-To-Use
AMS 3046 Magnetic Particles, Fluorescent, Wet Method, Oil Vehicle, Aerosol Packaged
AMS 5062 Steel, Low Carbon Bars, Forgings, Tubing, Sheet, Strip, and Plate 0.25 Carbon, Maximum
AMS I-83387 Inspection Process, Magnetic Rubber
AS 4792 Water Conditioning Agents for Aqueous Magnetic Particle Inspection
AS 5371 Reference Standards Notched Shims for Magnetic Particle Inspection

2.5 *Federal Standards:*[5,6]
FED-STD-313 Material Safety Data Sheets, Preparation and the Submission of

2.6 *Military Standards:*[5,7]
A-A-59230 Fluid, Magnetic Particle Inspection, Suspension

2.7 *OSHA Document:*[5,8]
29 CFR 1910.1200 Hazard Communication

2.8 *ISO Documents:*[5,9]
ISO 7810 Identification Cards—Physical Characteristics
ISO 9712 Nondestructive Testing—Qualification and Certification of NDT Personnel

4.2 This practice establishes the basic parameters for controlling the application of the magnetic particle testing method. This practice is written so that it can be specified on the engineering drawing, specification, or contract. It is not a detailed how-to procedure to be used by the examination personnel and, therefore, must be supplemented by a detailed written procedure that conforms to the requirements of this practice.

**5. General Practice**

5.1 *Personnel Qualification*—Personnel performing examinations in accordance with this practice shall be qualified and certified in accordance with ASNT Recommended Practice No. SNT-TC-1A, ANSI/ASNT Standard CP-189, ISO 9712, or as specified in the contract or purchase order.

5.2 *Agency Qualification*—If specified in the contractual agreement, NDT agencies shall be qualified and evaluated as described in Specification E543. The applicable edition of Specification E543 shall be specified in the contractual agreement.

5.3 *Written Procedure*—Magnetic particle testing shall be

# CERTIFICACIÓN DEL PERSONAL

- Se deben entregar los problemas resueltos según numero de grupo.
- Se deberá hacer un informe de ensayo para cada uno de los ensayos realizados.
  - Debe estar toda la información que se considera relevante
  - Debe existir un resultado como conclusión
  - Formato adecuado

# ¿QUÉ SE DEBE ENTREGAR COMO TP?